FE SEM 2 Rev NEP 2020 Syllabus Scheme for
Elective Chemistry

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- ENGINEERING MATERIALS -

MODULE IV : Plastics and Elastomers

(No. of hours – 05, Weightage of marks with option – 16)

- **Course Objective**: To learn important types, synthesis and uses of plastics and elastomers.
- **Course Outcome**: The students will be able to Utilize different plastics and elastomers in industries based on their properties.
- **Prerequisite**: Knowledge of basic properties of polymers.

MODULE IV : Plastics and Elastomers

Sr. No.	TOPIC	REMARKS
1.	A) Introduction to Plastics - Thermoplastic and Thermosetting plastics, compounding of plastics, Application of Plastics, Numerical based on Degree of polymerization, Density and mass, tensile strength of polymer	• Difference between Thermoplastics & Thermosetting plastics • Minimum one function & one example of each ingredient in compounding of plastic • Numericals on calculation of Degree of polymerisation, Density and mass, tensile strength of polymer Sample numericals will be provided.
2.	B) Introduction to elastomers - structural requirement of elastomer, natural rubber, processing of natural rubber, drawbacks, compounding of rubber	• Structure of Natural rubber • Manufacturing process of Natural rubber • Drawbacks of natural rubber, vulcanisation of rubber • Minimum one function & one example of each ingredient in compounding of rubber (including vulcanizing agents)

Sr. No.	TOPIC	REMARKS
2.	C) Synthesis of commercial polymers: i) Plastics: Preparation, properties and uses of a) Polymethyl Methacrylate (PMMA), b) Polytetrafluoroethylene (PTFE)	• Synthesis with reaction, 3-4 properties and 3-4 uses of each polymer
	ii) Elastomers: Preparation, properties and uses of a) Polyurethane Rubber, b) Silicone rubber	• Synthesis with reaction, 3-4 properties and 3-4 uses of each polymer

Reference Books (Plastics and Elastomers):

1. Engineering Chemistry - Jain and Jain, Dhanpat Rai Publication.
2. A textbook of Engineering Chemistry by S. S. Dara, S. Chand and Company.
3. Textbook of Polymer science: F.W. Billmeyer
4. Fundamentals of Polymer science & Engineering- Anilkumar & S K Gupta, Tata McGraw Hill, New Delhi

Plastics

The word plastic is derived from Greek word “Plastikos” i.e. fit for moulding. Plastics are high molecular weight organic material, which can be moulded into desired stable shapes by application of heat and pressure.

Thus, **Plastics are defined as an artificial material generally of synthetic organic origin having high molecular weight and can be shaped by applying heat and pressure often in a mould.**

In other words, plastics are nothing but the material, which shows the property plasticity. Plasticity is a property by means of which a material undergoes permanent deformation under stress.

Difference between thermoplastics and thermosetting plastics

S.N.	Thermoplastic Plastics	Thermosetting Plastics
1	They are formed by addition polymerisation	They are formed by condensation polymerisation
2	They are linear polymers with negligible branching	They form 3D network
3	They soften on heating	They do not become soft on heating
4	Soft, weak and less brittle	Hard, strong and brittle
5	They are soluble in organic solvents	They are insoluble in almost all organic solvents
6	They have relatively low molecular weight	They have relatively high molecular weight

Compounding of Plastic

Resins/Binders:

- It is a major part of the plastic. A classification of plastic is usually done according to the type of resin used for its manufacture.
- It also determines the type of treatment needed to mould the plastic.
- The resins used to prepare thermoplastics are called thermoplastic resins and the resins used to prepare thermosetting plastics are called thermosetting resins. Examples – PE, PVC, PVA (Thermoplastic resins), Phenol Formaldehyde, Urea Formaldehyde (Thermosetting resins)
- The main purpose of binder as the name implies is to hold all the other constituents of plastic together.

Compounding of Plastic

Fillers/Extenders:

- Fillers are added to the plastic to reduce the cost of the plastic per unit weight and to impart certain specific properties like hardness, T.S. toughness to the plastic.
- Examples of fillers -

i. Barium salts: To make the plastic impervious to X rays.

ii. Quartz & Mica: To improve hardness

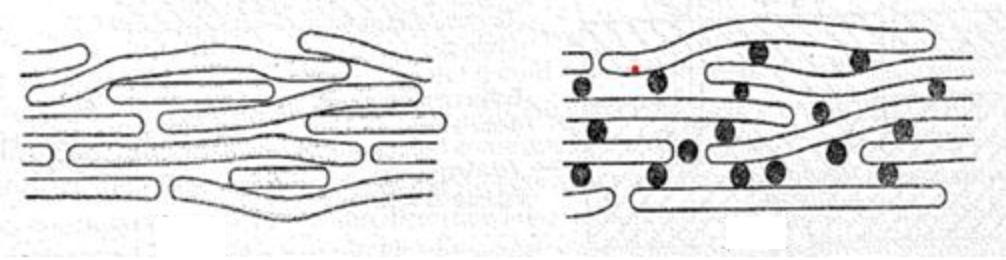
iii. Asbestos: To improve heat and corrosion resistance

- In addition to these, other materials used as fillers includes Cotton fibre, graphite, clay, paper pulp, metal oxide (ZnO, PbO), metal powder (Fe, Cu, Pd, Al)
- The fillers which enhance mechanical strength are called reinforcing fillers. For example, C black, cellulose fillers like cotton floc, wood flour.

Compounding of Plastic

Plasticizers:

- Plasticizers are the compounds which improves the property plasticity and flexibility of the plastic and reduces the temperature and pressure required for moulding.
- Their plasticizing property is mainly due to partial neutralization of intermolecular forces of attraction in the resin molecules. Thus, they impart greater freedom of movement between polymeric molecules of resin.
- Plasticizers are often used with thermo-softening plastics.
- Examples – Veg. oil, Tributyl phosphate, Tri-phenyl phosphate, camphor, esters of oleic, stearic or phthalic acid.



Compounding of Plastic

Lubricants:

- They imparts glossy finish to products
- They prevent sticking of plastic to the mould
- Examples: waxes, oleates, soaps, etc.

Dyes and pigments:

- Colour appeal is often of prime importance in high polymer artifacts
- The main colouring materials are organic dyestuffs and opaque inorganic pigments.

Stabilizers:

- In order to prevent the decomposition and discolouration of the plastic at the moulding temperature, stabilisers can be used

- Examples: $PbCrO_4$, stearates of Pb, Ca, Ba, etc.

Compounding of Plastic

Catalyst:

- Catalyst is used only in case of thermosetting plastic to accelerate the condensation polymerisation of fusible resins during moulding into infusible form.
- Thus, it reduces time required for curing the plastic article during moulding.
- Examples of catalyst are H_2O_2, $(C_6H_5CO)_2O$, Metals like Ag, Cu & Pb

Applications of Plastics

1. For making electrical goods and electrical appliances such as plugs, switches, holders, Radio T.V. cabinets, telephone parts, electrical cables.

2. In Aeronautical Engineering, For making furniture.

4. For making handles of tools and covers of machines.

5. For making floor and wall linings.

6. For heat and sound insulation in cold storage, refrigeration, theatres and auditorium.

7. For making hoses, pipes, tubes, tanks and tank linings for storing chemicals.

8. For making household articles like combs, toys, trays, containers, toilet and bathroom goods, toothbrush, bottles.

Numerical Problems

Problem 1: Density and Mass Calculation

Problem:

A plastic component is made of thermoplastic Polyvinyl Chloride (PVC) with a density of 1.4 g/cm^3. If the component has a volume of 250 cm^3, calculate its mass.

Solution:

$$\text{Mass} = \text{Density} \times \text{Volume}$$

$$\text{Mass} = 1.4\,\text{g/cm}^3 \times 250\,\text{cm}^3$$

$$\text{Mass} = 350\,\text{g}$$

Numerical Problems

- Tensile strength is the maximum amount of tensile stress a material can withstand before breaking, while Tensile stress is the force per unit area that stretches a material.

Numerical Problems

Problem 2: Tensile Strength of a Thermosetting Plastic

Problem:

A sheet of thermosetting plastic, Bakelite, has a tensile strength of 80 MPa. If a force of 4000 N is applied to a sample with a cross-sectional area of 0.5 cm^2, determine if the material will withstand the force without breaking.

Solution:

$$\text{Tensile Stress} = \frac{\text{Force}}{\text{Cross-sectional Area}}$$

$$\text{Tensile Stress} = \frac{4000\,\text{N}}{0.5\,\text{cm}^2}$$

Convert cm^2 to m^2:

$$0.5\,\text{cm}^2 = 0.5 \times 10^{-4}\,\text{m}^2$$

$$\text{Tensile Stress} = \frac{4000\,\text{N}}{0.5\times10^{-4}\,\text{m}^2}$$

$$\text{Tensile Stress} = 8 \times 10^7\,\text{N/m}^2$$

$$\text{Tensile Stress} = 80\,\text{MPa}$$

The tensile stress is 80 MPa, which matches the tensile strength of Bakelite. Therefore, the material will withstand the force without breaking.

Problem : Degree of Polymerization in Polyaniline (PANI)

Problem:

A sample of polyaniline (PANI) has a molecular weight of 20,000 g/mol. If the repeating unit of aniline ($C_6H_5NH_2$) has a molecular weight of 93 g/mol, calculate the degree of polymerization (DP) of the PANI.

Solution:

The degree of polymerization (DP) is defined as the ratio of the molecular weight of the polymer to the molecular weight of the repeating unit.

$$\mathrm{DP} = \frac{\text{Molecular weight of the polymer}}{\text{Molecular weight of the repeating unit}}$$

Given:

- Molecular weight of PANI = 20,000 g/mol
- Molecular weight of aniline repeating unit = 93 g/mol

$$\mathrm{DP} = \frac{20{,}000\ \mathrm{g/mol}}{93\ \mathrm{g/mol}}$$
$$\mathrm{DP} \approx 215$$

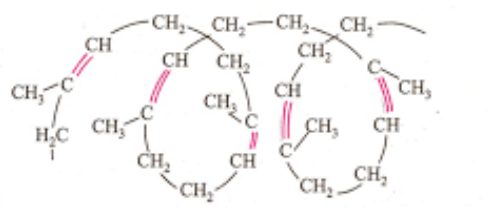
The degree of polymerization (DP) of the PANI is approximately 215.

Elastomers

• Structure of Natural rubber:

1. Rubbers are also known as elastomers are high polymers, which have elastic properties in excess of 300 percent.
2. The elastic deformation in an elastomer arises from the fact that in the unstressed condition, an elastomer molecule is *not straight chained* but in the form of a **coil** and consequently, it can be stretched like a spring.
3. <u>The unstretched rubber is amorphous</u>.
4. As stretching is done, the macromolecules get partially aligned with respect to each another, thereby causing crystallization and consequent, stiffening of material (due to increased attractive forces between these molecules).

The cis-polyisoprene molecules are in coiled or spring form



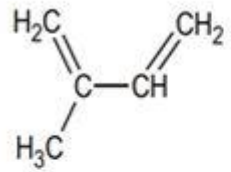
5. On releasing the deforming stress, the chains get reverted back to their original coiled state and the material again becomes amorphous.

Elastomers

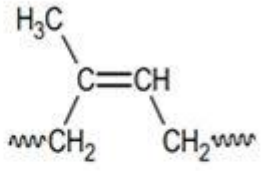
- Structure of Natural rubber:

6. Natural rubber consists of basic material latex, which is dispersion of isoprene.

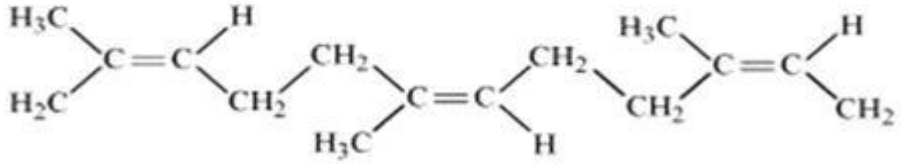
7. During the treatment, these isoprene molecules polymerize to form, long-coiled chains of cis-polyisoprene.



isoprene



cis-1,4 polyisoprene (natural rubber)



Natural rubber

8. The molecular weight of raw rubber is 100,000-150,000.

Manufacturing process of Natural rubber

1. The rubber latex is obtained by making incisions in the bark of the rubber trees and allowing the saps to flow out into small vessels.
2. Tapping is, usually, done at intervals of about 6 months.
3. The latex is emptied in buckets and transferred to a factory for treatment.

Processing of latex:

1. Latex is diluted to contain 15-20% of rubber and filtered to eliminate any dirt present in it.
2. Acetic or formic acid is then added, when rubber is coagulated to soft white mass. The coagulated mass (coagulum) is washed. The coagulated is treated as follows:

(a) Crepe rubber: The coagulum is allowed to drain for about 2 hours. Then it is passed through a creeping machine, which consists essentially of two rollers with longitudinal grooves, upon which water is sprayed. The spongy coagulum is covered into a sheet, which possesses an uneven rough surface resembling crepe rubber.

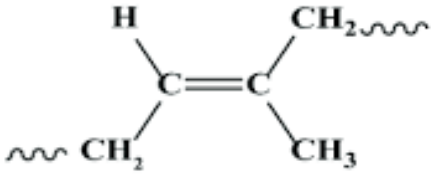
Manufacturing process of Natural rubber

(b) Smoked rubber:

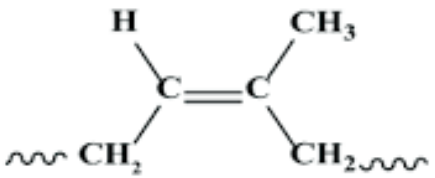
1. Coagulation is carried out in long tanks having sides with vertical grooves fitted with metal plates.
2. Diluted latex is poured into the tanks, with plates removed.
3. Then dilute formic acid is added and the mixture is stirred thoroughly.
4. The partition plates are then inserted into the grooves and tanks are kept undisturbed for about 16 hours.
5. The slabs are then passed through rollers, which gave ribbed pattern on the rubber sheets.
6. The sheets are then hung for about 4 days.
7. The crepe rubber thus obtained in ambre in colour and translucent.

Gutta percha

1. It is obtained from the mature leaves of dichopsis gutta and palagum gutta trees.
2. Gutta percha may be recovered by solvent extraction, when insoluble resin and gums are separated.



Gutta-percha (*trans*-1,4-polyisoprene)



Natural rubber (*cis*-1,4-polyisoprene)

3. At room temperature, gutta percha is tough, but it softens and becomes tacky at about 1000 ^{0}C. It is soluble in aliphatic hydrocarbons, but insoluble in aromatic and chlorinated hydrocarbons.
4. Structurally, its is trans-isoprene; where natural rubber is cis-isoprene.
5. **Uses**: In the manufacture of golf-ball covers, submarine cables, adhesives, and tissues for surgical purposes.

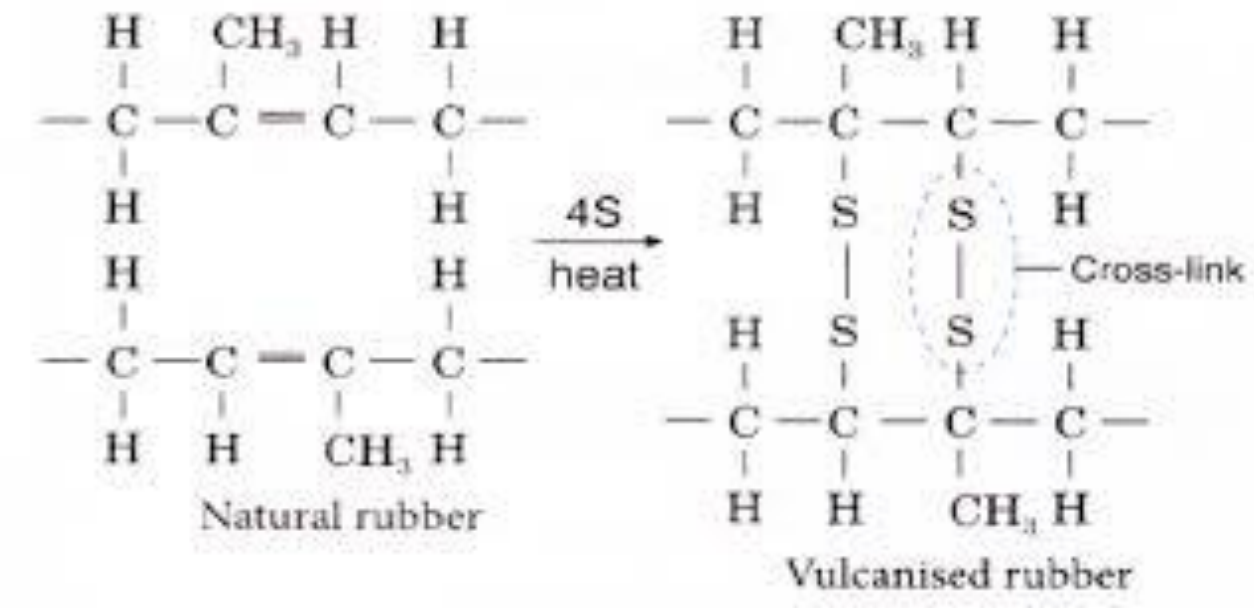
Drawbacks of natural rubber

1. It is plastic in nature.
2. It is weak.
3. It has large water absorption capacity.
4. It is non-resistant to non-polar solvents like vegetable and mineral oils, gasoline and benzene and carbon tetrachloride.
5. It is attacked by oxidizing agents like nitric acid, conc. sulphuric acid, chromic acid, etc.
6. It perishes, due to oxidation in air.
7. It swells in organic solvents.
8. It possesses marked tackiness.
9. It has little durability.

Vulcanization of Rubber

1. It is discovered by Charles Goodyear in 1839.
2. To improve the properties of rubber, it is compounded with some chemicals like sulphur, hydrogen sulphide, benzoyl chloride, etc.
3. Most important is addition of sulphur.
4. The process consists in heating the raw rubber with sulphur 100-140 ^{0}C.
5. The added sulphur combines chemically at the double bonds of different rubber springs.
6. Vulcanization thus serves to stiffen the material by a sort of anchoring and consequently, preventing intermolecular movement of rubber springs.
7. The extent of stiffness of vulcanized rubber depends on the amount of sulphur added.
8. **For example**, a tyre rubber may contain 3-5% sulphur, but a battery case rubber may contain as much as 30% sulphur.

Vulcanization of Rubber



Advantages of Vulcanization

Vulcanized rubber:

1. It has good tensile strength and extensibility.
2. It has excellent resilience.
3. It possesses low water absorption tendency.
4. It has higher resistance to oxidation and to abrasion.
5. It has much higher resistance to wear and tear as compared to raw rubber.
6. It is better electrical insulator.
7. It is resistant to organic solvents.
8. It has only slight tackiness.

Compounding of rubber

Compounding is mixing of the raw rubber (synthetic or natural) with other substances so as to impart the product specific properties suitable for particular job.

Besides rubber, the following materials may be incorporated:

1. Softeners and plasticizers: are added to give the rubber greater tenacity and adhesion. Important materials are vegtables oils, waxes, stearic acid, rosin, etc.
2. Vulcanizing agents: the main substance added is sulphur.
3. Accelerators: these materials drastically shorten the time required for vulcanization. E.g. Benzothiozole and zinc alkyl zanthate.

Compounding of rubber

4. Antioxidants: Rubber has tendency to perish due to oxidation. For this reason, antioxidation materials, such as complex amines (like phenyl naphthylamine) and phosphates are added.

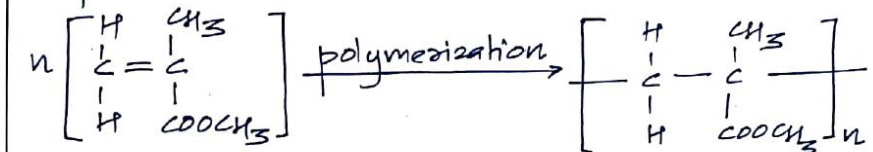
5. Reinforcing fillers: are added to give strength and rigidity to the rubber products. Common reagents are carbon black, zinc oxide, calcium carbonate.

6. Coloring matter: these are added to give the rubber product the desired colour. For white products, the usual pigment is titanium oxide; while for colored products, pigments (like chromium oxide-green, ferric oxide-red, antimony sulphide-crimson, lead chromate-yellow) and organic lake colours are added.

PMMA

2) Polymethyl Methacrylate (PMMA)

Preparation:-



Methyl Methacrylate (MMA)

Polymethyl Methacrylate (PMMA)

Properties:-

1) It becomes rubber-like at a temp. above 65°C.
2) It is hard, fairly rigid material.
3) It has high optical-transparency.
4) It has high resistance to sunlight.
5) It has low scratch resistance

Uses:-

1) For making lenses, optical parts of instruments air craft, light fixtures, artificial eyes, decorative articles, etc.
2) As it is more transparent than glass, it is used for making window glasses
3) It is found in paints. Acrylic 'latex' paints often contain PMMA suspended in wates.

PTFE

- It is obtained by polymerization of water-emulsion of tetrafluoro ethylene, under pressure in presence of benzoyl peroxide as catalyst.



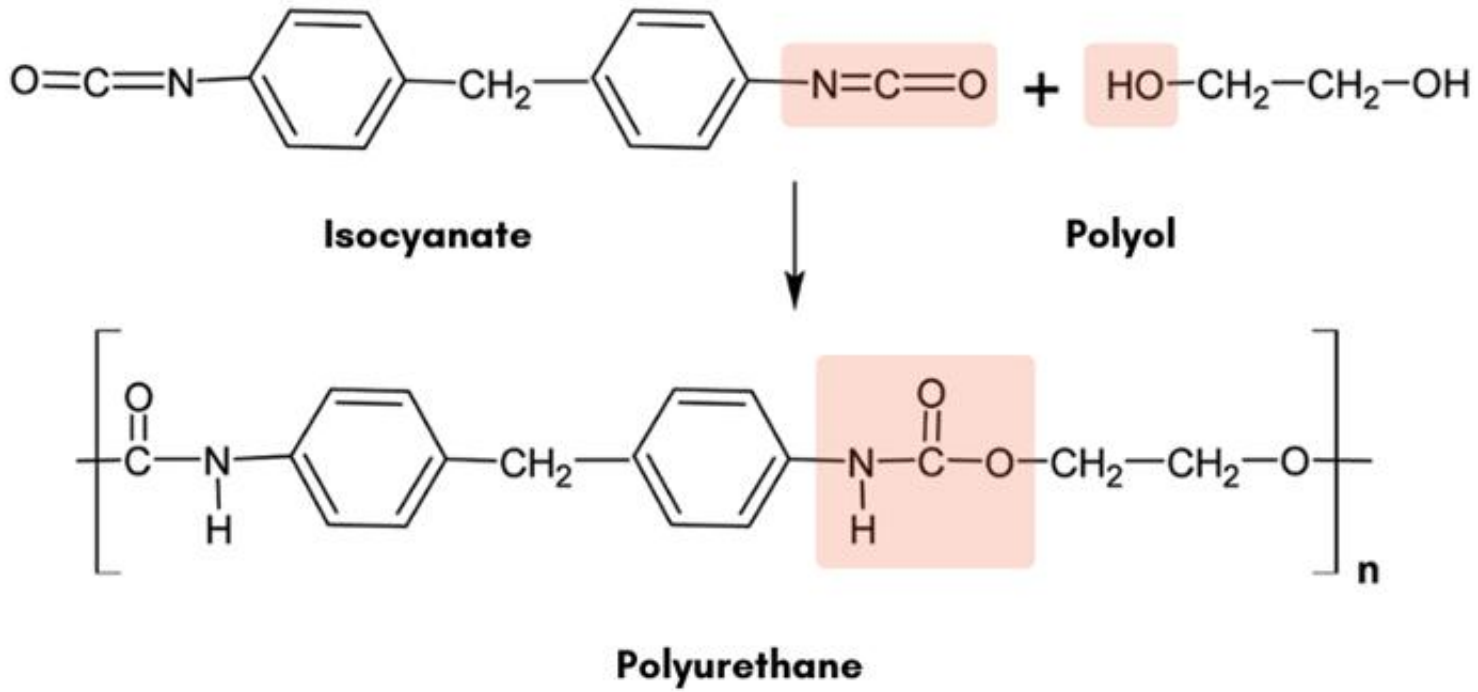
tetrafluoroethylene

polytetrafluoroethylene

- PTFE has extreme toughness, high softening point (about 350 ^{0}C), exceptionally high chemical resistance towards all chemicals (except hot alkali metal and hot fluorine), high density, waxy touch, very low coefficient of friction.
- Uses: 1. As an insulating material (for motors, transformers, cables, wires, fittings, etc.). 2. It is used for making gaskets, packings, pump parts, tank linings, chemical-carrying pipes, tubings and tanks, etc. 3. For coating and impregnating glass fibres, asbestos, fibres and cloths.

POLYURETHANE RUBBER (ISOCYANTE RUBBER)

- It is produced by reacting polyalcohols with di-isocyanates.



POLYURETHANE RUBBER (ISOCYANTE RUBBER)

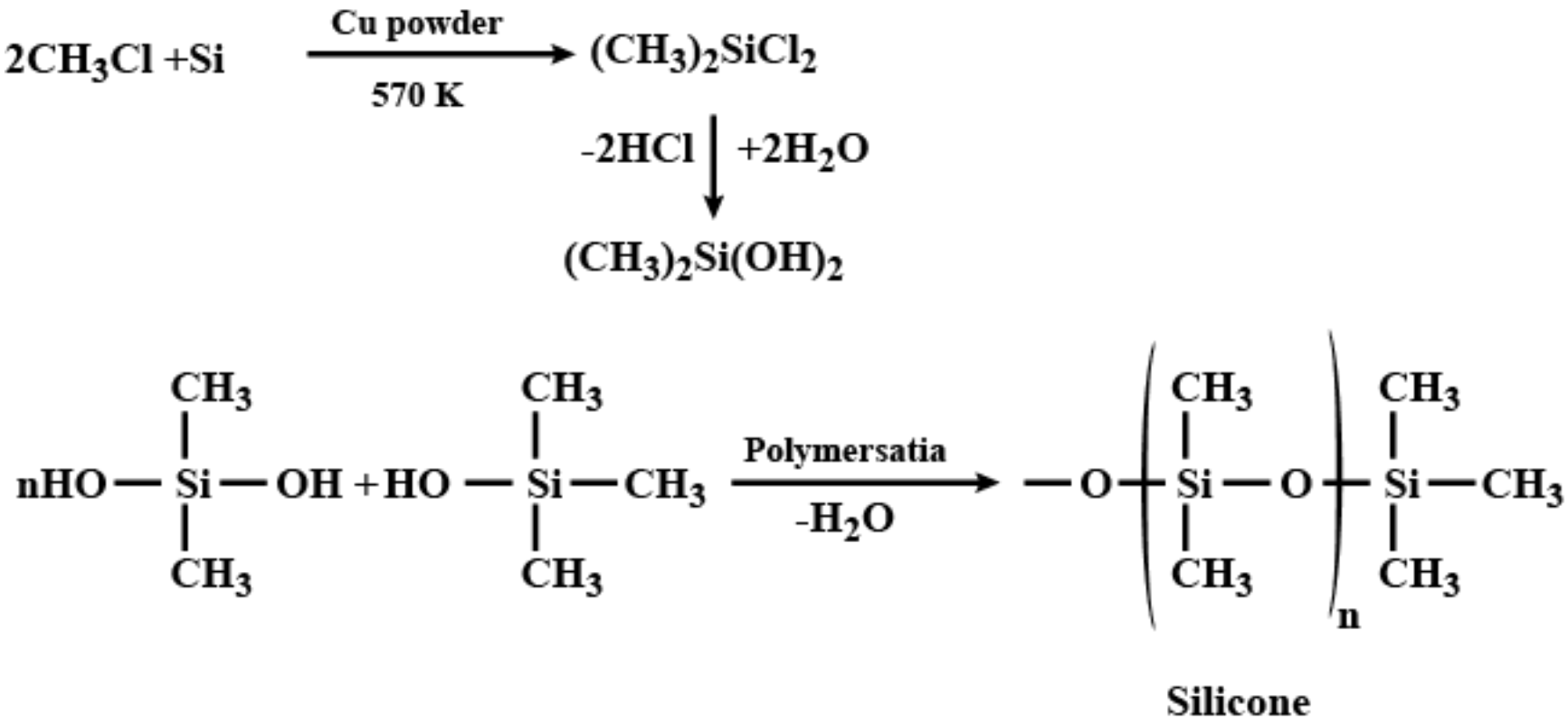
- Properties:

1. Polyurethanes are highly resistant to oxidation, because of their saturated character.
2. They show good resistance to many organic solvent, but are attacked by acids and alkalis, especially concentrated and hot.
3. The polyurethanes foams are light, tough and resistant to heat, abrasion, chemicals and weathering.

- Uses:
- For surface coatings and manufacture of foams and spandex fibres (85% Polyurethane + 15% Nylon/Polyester).

SILICONE RUBBER

- Preparation:



SILICONE RUBBER

- Properties:

1. They possess exceptional resistance to prolonged exposure to sunlight, weathering, most common oils, boiling water, dilute acids and alkalis.
2. They remain flexible in the temperature range of 90-250 ^{0}C and hence, find use in making tyres of fighter aircrafts, since they prevent damage on landing.
3. Silicone rubber at very high temperatures decomposes leaving behind the non-conducting silica (SiO_2) instead of coal tar.

- Uses:

1. As a sealing material in search-lights and in aircraft engines.
2. For insulating electrical wiring in ships.
3. In making lubricants, paints and protective coatings for fabric finishing and water-proofing.
4. As an adhesive in electronics industry.